

Synopsys: student guide

Start Fusion Compiler and Custom Compiler using the shared course environment.

1. Connect to your assigned server

On Windows, use MobaXterm with its X server running and SSH X forwarding enabled. On macOS, start XQuartz and connect with `ssh -Y`. Use the server assigned by your course; off-campus access may require the College remote-access service.

```
ssh -Y <username>@<assigned-server>
```

Replace both placeholders with your own username and assigned server address. Graphical tools need X forwarding.

2. Work in your own directory

```
mkdir -p ~/Labs/Lab_1  
cd ~/Labs/Lab_1
```

Use a separate directory for each lab or project. Designs, reports, logs and generated files belong in your own directories.

3. Load the environment

```
synopsys tsmc65LP
```

Run this once in each new terminal before launching the tools. It starts a configured shell with the TSMC 65nm kit. You do not need to clone the setup repository, copy libraries, or use `sudo`.

4. Check the shared libraries

```
checkndm
```

Run this during initial setup, or when asked to check the environment. The output should end with PASS. A long library report before PASS is normal. If it reports FAIL or missing libraries, contact the lab maintainer.

5. Launch your application

```
fc_shell -gui
```

Fusion Compiler graphical interface. For a command-line-only session, use `fc_shell`.

```
custom
```

Custom Compiler graphical interface. Use the custom command because it prepares the current directory before launching the application.

Everyday use and quick fixes

Keep this page handy during the labs.

Typical start for a new terminal

```
cd ~/Labs/Lab_1
synopsys tsmc65LP
fc_shell -gui
```

For Custom Compiler, use custom as the final command instead. To run both independently, open a second terminal and load the environment there too.

Leaving the environment

Save your work and close the application. At the configured operating-system shell prompt, run:

```
exit
```

This returns you to the shell you started from. If you are at the Fusion Compiler prompt, exit leaves Fusion Compiler first; it does not also unload the surrounding shell environment.

If something does not work

synopsys: command not found. Check that you are on the assigned server, then try a fresh login. In an existing tcsh session, run rehash. If the command is still missing, contact the maintainer.

checkndm or a tool is not found. Make sure you ran synopsys tsmc65LP in this terminal. If setup prints NOT ON PATH for a required tool, report that output to the maintainer.

A PDK is already loaded. Continue using the current environment. To switch kits, exit the configured shell and then load the desired kit.

No graphical window opens. Confirm your local X server is running and your SSH session has X forwarding enabled. In the remote terminal, echo \$DISPLAY should show a value. Reconnect with X forwarding if needed.

Library check fails or licence error appears. Send the error text and server name to the lab maintainer. Students do not need to rebuild the shared libraries or change licence settings.

Commands at a glance

synopsys	List available kits
synopsys tsmc65LP	Load the course environment
checkndm	Check shared reference libraries
fc_shell -gui	Fusion Compiler GUI
fc_shell	Fusion Compiler command line
custom	Custom Compiler GUI
exit	Leave the current shell/tool

The shared setup provides the applications and libraries. Follow the individual lab instructions to create or open a design, load scripts, and run the design flow.